HANNAH SKY GONZALEZ

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EDUCATION

University of California, San Diego Expected Graduation: Spring 2027

Bachelor of Science, Data Science & Bachelor of Arts, International Business

Coursework: Systems for Scalable Analytics, Probabilistic Modeling & Machine Learning, Representation Learning, Data Structures & Algorithms, Data Management, Data Analysis & Inference, Econometrics, Product Marketing & Management.

Activities: Women in Business, Recreational Basketball, Mentorship, and Toastmasters.

EXPERIENCE

JPMorgan Chase & Co. June 2026 - Aug. 2026

Data Science Intern *Jersey City, NJ*

- Built a machine learning model utilizing **GPT** and **LightGBM** that mapped new and updating regulatory text to impacted teams, optimizing the manual review process for the compliance department.
- Developed a candidate retrieval system using **Word2Vec** embeddings and **cosine similarity** to filter thousands of regulatory records, ensuring scalable mapping across **16** business lines and **247** sub-lines.
- Achieved **90%** recall and **97%** peak accuracy across business lines through adaptive threshold-calibration and a **LightGBM** reranker, ensuring complete coverage for human reviewers.
- Overcame extreme data sparsity by engineering candidate generation and global fallback mechanisms, delivering usable model recommendations for business lines with fewer than **10** historical examples.

UC San Diego Halıcıoğlu Data Science Institute Nov. 2025 - Present

Lab 3.0 Fellow *San Diego, CA*

- Taught **Arduino** and data science curricula to over **100** K-12 students, expanding STEM accessibility for underrepresented communities at a predominantly Latinx middle school.
- Engineered interactive robotics projects integrating **servos** and **LEDs** to translate complex data and hardware concepts into accessible art workshops for diverse skill levels.
- Scaled program outreach by building a targeted database of **200+** school administrators, driving lab partnerships and increasing educational access across the county.

LOZO June 2025 - Aug. 2025

Research Intern, Product Management *Shanghai, China*

- Led usability testing for an **AI**-powered speech analysis web platform, evaluating the end-to-end user experience across K-12 educators, students, and parents.
- Synthesized qualitative user feedback into distinct personas and journey maps, identifying key friction points and feature gaps within the different user portals.
- Delivered actionable product recommendations that simplified the **AI** evaluation workflow, ensuring the speech and debate platform was intuitive and engaging for all user segments.

PROJECTS

NCAA March Madness Cinderella Anomaly Detection

- Predicted tournament upsets to flag undervalued teams, using **Logistic Regression**, **SMOTE**, and **PCA** to boost F1 **187%**.
- Ensured reliable scouting predictions by preventing data leakage through strict chronological testing and StratifiedGroupKFold cross-validation.
- Deployed a serverless **Vercel** web app, enabling sports analysts to run real-time game predictions with zero latency.

League of Legends Position Prediction

- Built a **Random Forest** model predicting a player's position based on gameplay metrics with **77.3%** accuracy.
- Processed a 19k+ dataset through an **ETL** pipeline, using **Matplotlib** visualizations to refine model features.
- Validated model fairness across player segments using statistical hypothesis testing to ensure unbiased predictions.

Twitter Student Sentiment Analysis

- Analyzed **1.6M** tweets using **Pandas**, detecting a **25%** spike in negative community sentiment during finals week.
- Validated models via **SciPy** statistical testing and built interactive **Plotly** dashboards to visualize actionable sentiment trends.

SKILLS

Languages & Tools: Python, SQL, R, Java, HTML/CSS, JavaScript, Git, GitHub, Jupyter Notebook.

Data Science & ML: Pytorch, Pyspark, Dask, Pandas, NumPy, SciPy, scikit-learn, Matplotlib, ML Algorithms.

Big Data & Cloud: AWS, Apache Spark, MapReduce, Large-scale Data Management, Memory-Scalable Analytics.

AI & Developer Tools: GenAI (ChatGPT, Gemini, Claude), GitHub Copilot, Vercel, OracleDB.